VINEYARD
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DRINKING WATER IMPACT FEE FACILITY PLAN

(HAL Project No.: 319.04.100)

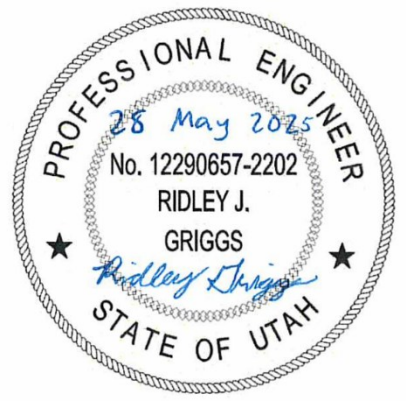
Adopted May 28, 2025

HANSEN ALLEN & LUCE inc
ENGINEERS

VINEYARD CITY

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Ridley J. Griggs, P.E.
Project Manager

HANSEN ALLEN & LUCE INC
ENGINEERS

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IMPACT FEE CERTIFICATION

The Utah Impact Fee Act requires certifications for the Impact Fee Facilities Plan (IFFP). Hansen, Allen & Luce provides these certifications with the understanding that the recommendations in the IFFP are followed by City Staff and elected officials. If all or a portion of the IFFP is modified or amended, or if assumptions presented in this analysis change substantially, this certification is no longer valid. All information provided to Hansen, Allen & Luce, Inc. is assumed to be correct, complete, and accurate.

IFFP Certification

Hansen, Allen & Luce, Inc. certifies that the Impact Fee Facilities Plan (IFFP) prepared for the drinking water system:

1. includes only the costs of public facilities that are:
    a. allowed under the Impact Fees Act; and
    b. actually incurred; or
    c. projected to be incurred or encumbered within six years after the day on which each impact fee is paid;
2. does not include:
    a. costs of operation and maintenance of public facilities;
    b. costs for qualifying public facilities that will raise the level of service for the facilities, through impact fees, above the level of service that is supported by existing residents;
    c. an expense for overhead, unless the expense is calculated pursuant to a methodology that is consistent with generally accepted cost accounting practices and the methodological standards set forth by the federal Office of Management and Budget for federal grant reimbursement; and
3. complies in each and every relevant respect with the Impact Fees Act.

HANSEN, ALLEN & LUCE, INC.

CHAPTER 1 INTRODUCTION

PURPOSE AND SCOPE

The City of Vineyard is located in north-central Utah County on the eastern shore of Utah Lake. The City is experiencing growth. To ensure availability of funds for growth-related infrastructure projects, an Impact Fee Facility Plan (IFFP) was commissioned by the City.

This report identifies those items that the Utah Impact Fees Act specifically requires, including demands placed upon existing facilities by new development and the proposed means by which the municipality will meet those demands.

IMPACT FEE COLLECTION

An impact fee is a one-time charge on new development to pay for that portion of a public facility that is required to support that new development. Impact fees enable local governments to finance public facility improvements necessary for growth, without burdening existing customers with costs that are exclusively attributable to growth.

In order to determine the appropriate impact fee, the cost of the facilities associated with future development must be proportionately distributed. As a guideline in determining the “proportionate share”, the fee must be found to be roughly proportionate and reasonably related to the impact caused by the new development.

CAPITAL FACILITIES PLAN

A drinking water system master plan was prepared in conjunction with this analysis and is incorporated by reference into this analysis.

The master plan for the City’s drinking water system is more comprehensive than the IFFP. It provides the basis for the IFFP and IFA and identifies all capital facilities required of the drinking water system for the 20-year planning range, including maintenance, repair, replacement, and growth-related projects. The recommendations made within the master plan are in compliance with current City policies and standard engineering practices.

A hydraulic model of the drinking water system was prepared to aid in the analyses performed to complete the drinking water system master plan. The model was used to assess existing performance, to establish a proposed level of service and to confirm the effectiveness of the proposed capital facility projects to maintain the proposed level of service over the next 10 years.

CHAPTER 2 SYSTEM DEMAND AND CAPACITY

GENERAL

The purpose of this section is to identify the current level of service, characterize the facilities of the existing system, and determine the remaining capacity of these facilities.

The existing drinking water system is comprised of a distribution network, water storage facilities, and water sources. These facilities are found within two pressure zones. Figure 2-1 illustrates the existing water system and its service area.

LEVEL OF SERVICE

A level of service (LOS) has been established for Vineyard City for the drinking water system. The level of service establishes the sizing criteria for the Area's distribution facilities, source facilities, storage facilities, and water rights. The level of service standards are shown below:

Level of Service

- Peak Day Demand: 1,248 gpd/ERC
- Indoor Storage Capacity (Equalization): 590 gallons/ERC
- Indoor Fire Storage Capacity: 58 gallons/ERC (0.6 MG fire storage requirement at buildout divided by the number of ERCs at buildout (10,260))
- Distribution Capacity: 40 psi minimum during peak day demand conditions, and 30 psi minimum during peak instantaneous conditions

Fire Suppression

- Minimum Fire Flow: 1,500 gpm for 2 hours
- Maximum Fire Flow: 2,500 gpm for 4 hours
- Fire Storage (buildout): 600,000 gallons
- Minimum Pressure: 20 psi residual during peak day + fire flow event

EXISTING AND FUTURE EQUIVALENT RESIDENTIAL CONNECTIONS

Water demands from non-residential water users, such as commercial, industrial, or civic water users have been determined in terms of an Equivalent Residential Connection (ERC). The use of ERCs is a common engineering practice used to describe the entire system's usage based on a common unit of measurement. An ERC is equal to the average demand of one average single-family residential connection in Vineyard City. Using ERCs for analysis is a way to allocate existing and future demands over non-residential land uses.

Vineyard operates a pressurized irrigation (PI) system that provides water service to certain institutional properties. It will also be expanded to some other properties in the future.

Table 2-1 includes a summary of ERCs for each pressure zone under existing conditions and at buildout.

Lindon (emergency)

1600 N

CUWCD 1600 N

Lindon (emergency)

800 N

CUWCD 800 N

Mill Rd

400 N

Main St

Center St

Holdaway Rd

Orem Center St

400 S

Orem 400 S (emergency)

Orem Sleepy Ridge (emergency)

N

0 1,500 3,000 6,000 Feet

Legend

Interconnections

- CUWCD
- Lindon
- Orem

Pipe Diameter

- 8-inch
- 10-inch
- 12-inch
- 14-inch

Pressure Zone

- North Pressure Zone
- South Pressure Zone

Date: 2/20/2024
Document Path: H:\Projects\319 - Vineyard\04.100 Drinking Water Master Plan\GIS\Working\Fig 2-1 Existing Drinking Water System.mxd

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VINEYARD DRINKING WATER MASTER PLAN

EXISTING DRINKING WATER SYSTEM

FIGURE 2-1

Table 2-1
Existing and Future Service by Pressure Zone

Pressure Zone	Existing ERCs	Buildout ERCs[1]
North	1,090	-
South	2,240	-
Total	**3,330**	**10,260**

1. At buildout, the system will have one pressure zone.

METHODOLOGY USED TO DETERMINE EXISTING SYSTEM CAPACITY

Each component of the drinking water system were assessed in terms of gallons per minute (for peak day source), acre-feet per year (for annual source), or gallons (for storage). Demands on each component were computed by applying the level of service to the number of ERCs. The difference between the capacity of the component and the demand on the component is the component's remaining capacity, which can be used to serve ERCs. A hydraulic model was developed for the purpose of assessing system operation and distribution capacity.

SOURCE CAPACITY

Source capacity in the drinking water system is currently provided through agreements with the City of Orem and Central Utah Water Conservancy District (CUWCD). Vineyard is currently working to secure additional source capacity with CUWCD and is planning to assess fees for this water separate from the impact fee. As such, source capacity is not covered in detail in this IFFP.

STORAGE CAPACITY

Storage capacity in the drinking water system is currently provided through agreements with the City of Orem and Central Utah Water Conservancy District (CUWCD). The City has secured long-term storage capacity in the CUWCD system. The storage agreement with the City of Orem is temporary. Storage capacity secured through long-term agreement is listed in Table 2-2. Capacity secured through temporary agreements is not listed.

Table 2-2
Existing Storage Capacity

Parameter	Storage (MG)
Long-Term Capacity[1]	2.00
System Requirement[2]	3.04
Difference	**-1.04**

1. Long-term storage capacity secured through agreements with CUWCD.
2. See Table 4-4 in the Drinking Water Master Plan

Currently, the City does not have long-term excess storage capacity secured through contract agreements but is meeting current requirements through a temporary agreement with Orem.

DISTRIBUTION SYSTEM

Pipe diameters range from 4 inches to 20 inches, with the majority being 8 inches in diameter. The function of the larger pipes in the system is to facilitate filling of the storage tanks and meet peak day and fire flow demands. Smaller pipes facilitate local distribution. Figure 2-1 in the Drinking Water Master Plan illustrates the existing distribution pipelines. A hydraulic model was used to identify future pipeline projects (see Chapter 3 for details). All pipeline projects were sized to meet projected buildout requirements. A summary of capacity remaining in the distribution system is summarized in Table 2-3.

Table 2-3
Capacity of Distribution Pipes

System Component	Capacity (ERCs)	Existing (ERCs)	Remaining Capacity (ERCs)
Distribution Pipes	10,260	3,330	6,930

CHAPTER 3 IMPACT FEE FACILITY PLAN

This section relies on the data presented in the previous sections to identify projects needed to support future growth, their planned capacity, and their estimated costs. Also included in this section are the possible revenue sources that the City may consider for funding the recommended projects.

GROWTH PROJECTIONS

The development of impact fees requires growth projections over the next ten years. Growth projections for Vineyard City were made by incorporating the growth rate presented in the Master Plan. Total growth projections for the City through the 10-year planning horizon are summarized in Table 3-1. Areas of projected growth are shown in Figure 3-1.

Table 3-1
Growth Projections

Year	ERCs
2024	3,330
2025	3,935
2026	4,549
2027	5,226
2028	5,637
2029	5,999
2030	6,361
2031	6,689
2032	7,080
2033	7,345
2034	7,610
10-year Difference	**+4,280**

The existing system served about 3,330 ERCs at the beginning of 2024. Projected growth adds 4,280 ERCs in the next 10 years for a total of 7,610 ERCs.

FUTURE STORAGE REQUIREMENTS

Projected storage requirements for Vineyard City are calculated based on anticipated growth in the next ten years and the LOS. Table 3-2 shows source requirements for future growth.

Table 3-2
10-Year Storage Requirements and Capacity

10-Year Growth (ERCs)	Level of Service (gal/ERC)	Requirement (MG)	Capacity Remaining[1] (MG)	Surplus/Deficit (MG)
4,280	590	2.53	-1.04	-3.57

1. See Table 2-2.

Table 3-3 indicates that the existing storage capacity is inadequate to meet demands in the 10-year window. Additional storage capacity is needed.

Impact fees for source and storage will be computed based on the cost of existing facilities and future projects needed to provide capacity within the 10-year planning window.

COST OF FUTURE FACILITIES

The facilities and costs presented in Table 3-3 and Table 3-4 and shown on Figure 9-1 in the Drinking Water Master Plan represent future projects that are impact fee eligible. These proposed projects are essential to maintain the current level of service while accommodating future growth within the next 10 years. The facility sizing for the future proposed projects was based on the proposed level of service with growth projections and hydraulic modeling. The proposed impact fee will be based on the projected cost of future construction projects. Detailed cost estimates for the impact fee-eligible projects are included in Appendix A.

There are no projects listed in the Drinking Water Master Plan which are 100% related to correcting existing deficiencies.

Cost estimates of the projects that are only partially impact fee eligible are described in Table 3-3.

Table 3-3
Estimated Cost of Future Facilities Partially Impact Fee Eligible

Project	Map ID	Type of Project	Total Cost	% Impact Fee Eligible	Total Impact Fee Eligible Cost
6.0 MG Storage Tank	1	Storage	$10,539,102	64%[1]	$6,746,015
Booster Pump Station	1	Distribution	$7,026,068	68%[2]	$4,745,678
Pipeline Projects (see Table 9-1 of the Drinking Water Master Plan)	1	Distribution	$14,270,267	68%[2]	$9,638,689
Total		**-**	**$31,835,437**	**-**	**$21,130,381**

1. Eligible cost is calculated as the remainder of the total number of existing ERCs (3,330) divided by the total number of ERCs (9,252) served from 6.0 MG storage tanks. Number of ERCs were found by dividing the storage capacity the LOS at buildout (Equalization: 590 gal/ERC & Fire Storage: 58 gal/ERC).
2. Impact fee eligible for the distribution facilities was conservatively estimated as the difference between the existing ERC count (3,330) and the projected ERC count at 2060 (10,260).

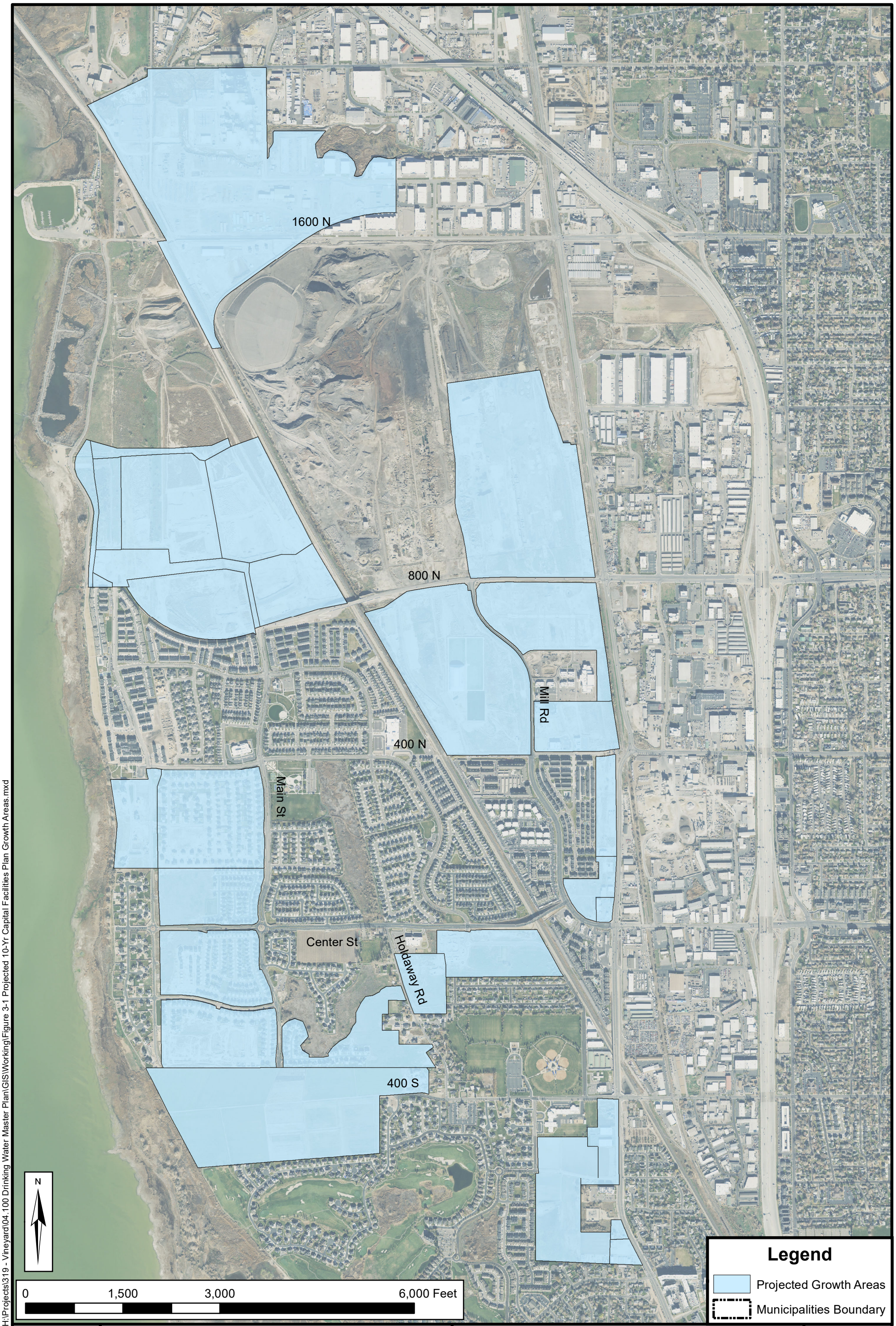
Cost estimates for the future projects shown in Table 3-4 are considered to be 100% eligible as they are only necessary to support future growth. Each of these projects is planned to be constructed within the next 10 years and are sized to provide capacity through buildout.

Table 3-4
Estimated Cost of Future Facilities Fully Impact Fee Eligible

Project	Map ID	Distribution	Storage	Total
Mill Road North Extension	2	$2,191,000	$0	**$2,191,000**
Pump Station Upgrade	3	$708,000	$0	**$708,000**
South Downtown Distribution	5	$2,191,000	$0	**$2,191,000**
Holdaway Farms Distribution	6	$911,000	$0	**$911,000**
East Geneva Distribution 1	7	$1,822,000	$0	**$1,822,000**
Total		**$7,823,000**	**$0**	**$7,823,000**

Planning

Planning services are also needed to support growth. The City intends to update their master plans approximately every 5 years and their impact fee studies approximately every 3 years. Considering this schedule, and the cost of the most recent impact fee updates, the ERCs served by each plan update were calculated as shown in Table 3-5.



Date: 4/30/2025
Document Path: H:\Projects\319 - Vineyard\04.100 Drinking Water Master Plan\GIS\Working\Figure 3-1 Projected 10-Yr Capital Facilities Plan Growth Areas.mxd

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VINEYARD CITY
DRINKING WATER SYSTEM

PROJECTED 10-YEAR GROWTH AREAS

FIGURE 3-1

Table 3-5
Planning Component of Impact Fee

Planning Document	Cost	% of Plan Associated with Growth[1]	ERCs Served[2]
2024 Drinking Water System Master Plan	$107,932.00	60%	1,896
2024 IFFP and IFA	$20,200.00	100%	2,669
Total	**$128,132.00**	**-**	**-**

1. Percentages to growth for the master plan was based on a review of the scope of the plan and associated fees for tasks associated with the existing system and future growth. The IFFP and IFA are 100% associated with growth.
2. ERCs served was defined as the amount of ERCs expected to develop during the 5-year life of the master plan and the 3-year life of the IFFP and IFA, respectively. See Table 3-1.

IMPACT FEES BY METER SIZE

Vineyard City assesses impact fees based on meter size. Vineyard considers a standard 3/4-inch meter as one ERC. Larger meters are assessed a capacity in ERCs based on American Water Works Association (AWWA) capacity factors. See Appendix B. Table 3-6 includes a summary of ERCs by meter size.

Table 3-6
ERCs by Meter Size

Meter Size	ERCs
3/4”	1
1”	1.67
1 ½”	3.33
2”	5.33
3”	10.00
4”	16.67
6”	33.33
8”	53.33
10”	76.67

NON-STANDARD IMPACT FEES

In cases where it is determined that the use of meter sizes to assess impact fees would not lead to an equitable result, ERCs may instead be calculated based on estimated peak day demand as follows:

$$\text{ERCs} = (\text{peak day demand, gpd}) \ / \ (1{,}248 \text{ gpd/ERC})$$

Examples where a meter size may not produce an equitable result include instances where a development uses multiple meters rather than a single meter or when the meter multiplier does not accurately reflect the expected multiplier of the customer as compared to a standard ERC.

REFERENCES

State of Utah. 2014c. Utah Code Annotated, Section Utah Code 11-36a: Impact Fees Act

Vineyard City. 2025. Hansen, Allen & Luce. Drinking Water System Master Plan

APPENDIX A

Estimated Project Costs

Vineyard City Capital Facility Plan
Drinking Water Recommended Improvements
Preliminary Engineers Cost Estimates

	Item	Unit	Unit Price	Quantity	Total Price
1	***Backbone Infrastructure Project***				
	6 MG Tank and Pump Station*	LS	$16,401,404.69	1	$ 16,401,405
	Pipelines*	LS	$13,753,752.52	1	$ 13,753,753
	* Actual Contract Costs			Engineering & Admin. (included above)	$ -
				Contingency (N/A)	$ -
				Total to Backbone Infrastructure Project	**$ 30,155,157**
2	***Mill Road North Extension***				
	16-inch diameter pipe	LF	$ 332	5500	$ 1,826,000
				Engineering & Admin. (10%)	$ 182,600
				Contingency (10%)	$ 182,600
				Total to Mill Road North Extension	**$ 2,191,000**
3	***Pump Station Upgrade***				
	400 hP pump	LS	$ 120,000	2	$ 240,000
	VFD and Conductors	LS	$ 100,000	2	$ 200,000
	Generator	LS	$ 150,000	1	$ 150,000
				Engineering & Admin. (10%)	$ 59,000
				Contingency (10%)	$ 59,000
				Total to Pump Station Upgrade	**$ 708,000**
4	***Storage Expansion***				
	Concrete Storage Tank	GAL	$ 1.50	3000000	$ 4,500,000
				Engineering & Admin. (10%)	$ 450,000
				Contingency (10%)	$ 450,000
				Total to Storage Expansion	**$ 5,400,000**
5	***South Downtown Distribution***				
	16-inch diameter pipe	LF	$ 332	5500	$ 1,826,000
				Engineering & Admin. (10%)	$ 182,600
				Contingency (10%)	$ 182,600
				Total to South Downtown Distribution	**$ 2,191,000**
6	***Holdaway Farms Distribution***				
	12-inch diameter pipe	LF	$ 292	2600	$ 759,200
				Engineering & Admin. (10%)	$ 75,920
				Contingency (10%)	$ 75,920
				Total to Holdaway Farms Distribution	**$ 911,000**
7	***East Geneva Distribution 1***				
	12-inch diameter pipe	LF	$ 292	5200	$ 1,518,400
				Engineering & Admin. (10%)	$ 151,840
				Contingency (10%)	$ 151,840
				Total to East Geneva Distribution 1	**$ 1,822,000**
8	***East Geneva Distribution 2***				
	12-inch diameter pipe	LF	$ 292	5600	$ 1,635,200
				Engineering & Admin. (10%)	$ 163,520
				Contingency (10%)	$ 163,520
				Total to East Geneva Distribution 2	**$ 1,962,000**
9	***North Downtown Distribution***				
	16-inch diameter pipe	LF	$ 332	3000	$ 996,000
				Engineering & Admin. (10%)	$ 99,600
				Contingency (10%)	$ 99,600
				Total to North Downtown Distribution	**$ 1,195,000**
10	***North Industrial Distribution***				
	12-inch diameter pipe	LF	$ 292	2600	$ 759,200
				Engineering & Admin. (10%)	$ 75,920
				Contingency (10%)	$ 75,920
				Total to North Industrial Distribution	**$ 911,000**
				Total Costs	**$ 47,446,000**

APPENDIX B

AWWA Meter Multipliers

Meter Size	AWWA (capacity)	Factor based on 3/4
5/8 inch	20	0.67
3/4 inch	30	1.00
1 inch	50	1.67
1-1/2 inch	100	3.33
2 inch	160	5.33
3 inch	300	10.00
4 inch	500	16.67
6 inch	1,000	33.33
8 inch	1,600	53.33
10 inch	2,300	76.67
12 inch	4,300	143.33
14 inch	6,400	213.33
16 inch	9,160	305.33
18 inch	14,650	488.33
20 inch	18,500	616.67